

From Theory to Model

Selection, restriction and specification in physical modelling

Paper IV of IV

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Abstract

Physics is frequently described as the study of laws, yet physicists rarely solve laws in the abstract. They construct and solve models. A theory supplies admissible quantities, structural relations, constitutive possibilities and interaction laws; a model selects which parts of that architecture are retained, restricts their behaviour to a tractable regime, and specifies the geometry, parameters and conditions of a particular problem. This essay develops a three-part account of the modelling act: *selection moves* determine the system, degrees of freedom and included interactions; *restriction moves* impose idealisations, approximations, constitutive choices and regimes; *specification moves* provide coordinates, gauge, geometry, parameters, boundary conditions and initial data. The framework is instantiated through three cases: a block on an inclined plane, a charged particle in prescribed electromagnetic fields, and a simple thermodynamic cycle. The analysis completes the four-paper programme by locating hinge pedagogy at the point where an otherwise open model still requires one further modelling decision.

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Physics deploys models

Physics is often presented as a progression from laws to dynamical equations to solutions of those equations of motion. The structural laws of mechanics, electromagnetism and thermodynamics have a reach over a wide set of phenomenology, yet a mass spectrometer, a railgun, a cyclotron and a charged droplet are distinct systems. Their different treatments do not lie primarily in the underlying theory rather in the models constructed from that theory.

A theory supplies a grammar: what kinds of quantities may appear, how they are related, which equations have structural, constitutive, dynamical or interactional roles, and what kinds of couplings are admissible. A model supplies an instantiation of that grammar: which *class* of objects are represented, what lies inside the system, which degrees of freedom are retained, which interactions are included or omitted, what regime is assumed, and what data makes the problem determinate.

The distinction may be written schematically as

$$\boxed{\text{theory}} + \boxed{\text{modelling decisions}} \longrightarrow \boxed{\text{model}} \longrightarrow \boxed{\text{predictions}}.$$

The central claim we make here is that the modelling decisions of a physicist falls into three functionally distinct groups:

$$\boxed{\text{selection}} \quad \boxed{\text{restriction}} \quad \boxed{\text{specification}}.$$

Selection determines what the model contains. Restriction determines how the retained components are permitted to behave. Specification determines which particular member of the resulting model family is being solved.

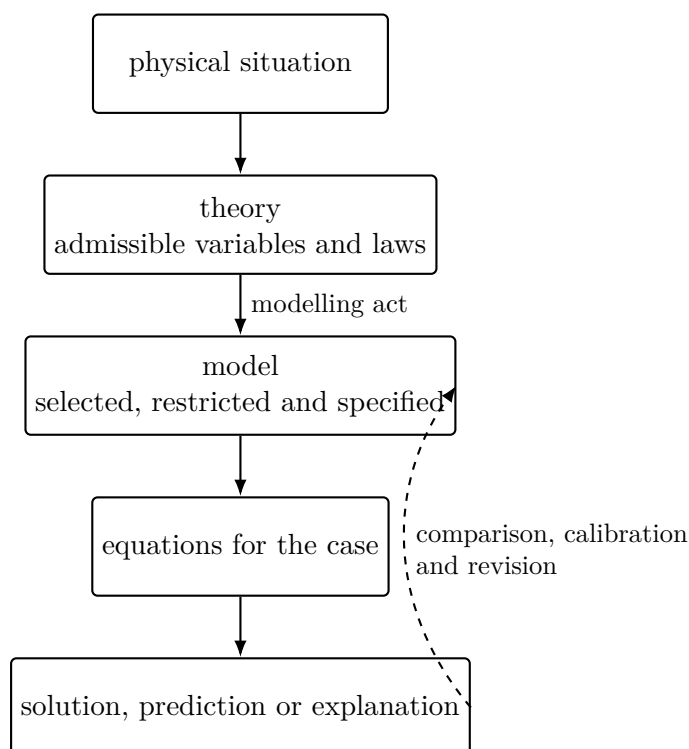


Figure 1: A theory is not applied to reality without mediation. The model is the constructed object through which the theory becomes calculable and testable.

Hestenes placed models at the centre of physics instruction, describing Newtonian mechanics as a system of “modelling games” directed toward the construction and validation of representations of

physical phenomena [4]. Halloun similarly defines models through composition, structure, domain and organisation, and treats modelling as involving selection, construction, validation, analysis and deployment [5]. The present proposal is narrower. It classifies the specific moves by which a theory is reduced to a solvable model.

This account sits between the architectural and epistemic analyses developed in Papers I and III. Paper I asked how a physical theory is organised while Paper III asked what kind of claim an individual equation makes. The present paper asks what happens between them: how an abstract theory is turned into a determinate physical model.

Models as mediators

The model is a distinct epistemic object and is more than a limited copy of a theory. Morgan and Morrison describe models as partially autonomous instruments that mediate between theories and the world [7]. Their autonomy matters because a model may contain assumptions, representations, parameters and idealisations that are neither direct statements of the underlying theory nor literal descriptions of the target system.

Nancy Cartwright's analysis of physical laws makes the same gap visible from another direction. Fundamental laws achieve explanatory scope partly because they abstract from the complications of actual situations; their application requires the construction of more local models in which *ceteris paribus* clauses, idealisations and causal compositions are made operative [6]. A law such as

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}$$

does not determine a trajectory until the system, forces, constitution, constraints and conditions have been supplied. And so this gives us a clean hierarchy:

Theory : What kinds of quantities and relations are admissible?

Model : Which instantiation applies here, under which restrictions?

Equation : Where does this line sit, and what kind of claim is it?

Teaching : Which module can be held fixed while another is varied?

Equations influence model construction; observations force revisions; pedagogical representations can make some modelling choices visible and hide others. So the hierarchy is not strictly linear and yet such distinctions usefully prevent a collapse in which theory, model, laws, equation and calculation are treated as interchangeable.

The three modelling parses

Constructing a physical model is often described simply as "making assumptions". That description is too coarse. Different assumptions perform different logical functions. Some determine what the model contains; others constrain how those entities behave; others merely identify the particular case to be solved.

I shall refer to these distinct modelling operations as *parses*. A parse is a decomposition of the modelling act according to the role each decision plays. Although these decisions are frequently made together in practice, they answer different questions and reduce different spaces of possibility.

Figure 2 summarises the framework. Beginning with an abstract physical theory, every model is constructed through three complementary parses:

- **Selection** — What entities, interactions and degrees of freedom are represented?
- **Restriction** — Under what constitutive assumptions, idealisations and regimes may they behave?
- **Specification** — Which particular geometry, parameters and conditions define the case being solved?

Together these transforms convert an abstract theory into a determinate model. The following sections examine each parse in turn.

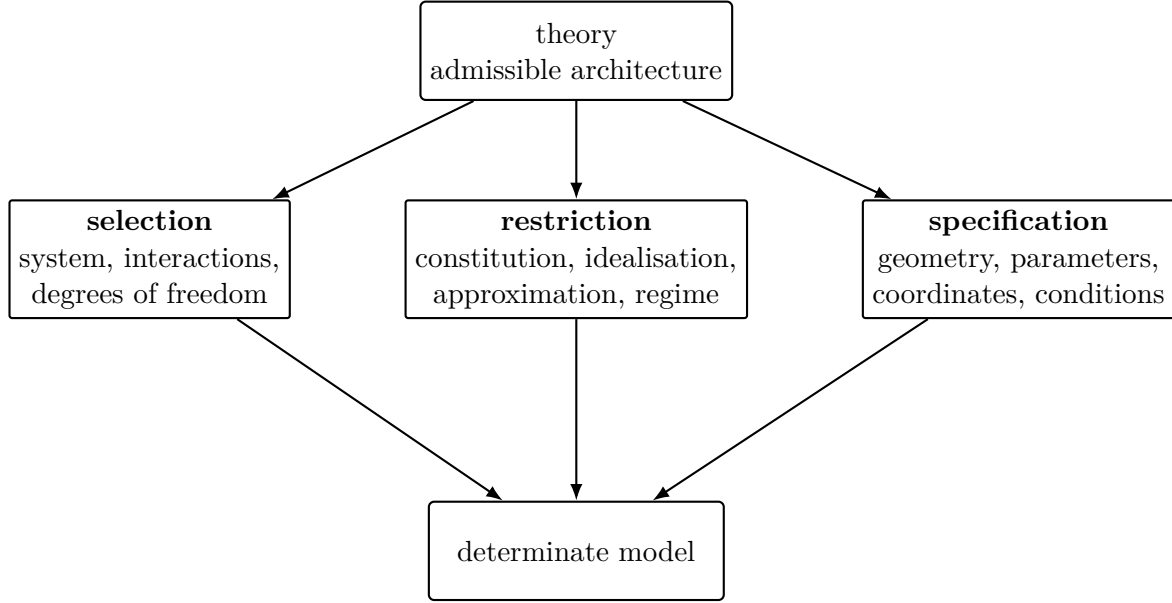


Figure 2: The three modelling moves. They are analytically distinct even when made together in practice.

Selection: what is included?

A selection parse determines the ontology of the working model. It fixes the system boundary, the retained degrees of freedom, the interactions represented explicitly, and those omitted or treated as external. Typical selection questions include:

- What is the system?
- Which bodies, fields or continua are represented?
- Which variables are solved for?
- Which interactions are included?
- Which influences are treated as external inputs?
- Which degrees of freedom are suppressed?
- Which symmetry class is retained?

A test-particle approximation is a selection move. It retains the particle and the action of a field upon it while excluding the particle’s back-reaction on the field. Treating a rope as a continuum rather than a chain of point masses is also a selection move because it changes the retained degrees of freedom. Assuming spherical symmetry and so removing angular dependence from the represented state space is another. Ignoring air resistance removes an interaction from the model that would otherwise require parameterized phenomenology.

Selection therefore answers:

What is this model about?

Selection also establishes the boundary across which interactions pass. In Paper I, the Lorentz force could appear as an external interaction for a particle in a prescribed field or as an internal coupling in a self-consistent particle–field model. The mathematics of the force law is unchanged but its placement changes because the model boundary changes.

Restriction: how may the retained system behave?

A restriction parse constrains the functional behaviour of what selection has retained. It includes *constitutive* choices, idealisations, approximations, regime assumptions and symmetry restrictions understood as constraints on admissible behaviour.

Typical restriction questions include:

- Is the response linear or nonlinear?
- Is the material isotropic or anisotropic?
- Is the motion non-relativistic?
- Is the field static, quasistatic or fully time-dependent?
- Is the body rigid, elastic or deformable?
- Is dissipation included?
- Which dimensionless quantities are assumed small?

The constitutive relation

$$\mathbf{D} = \epsilon \mathbf{E}$$

in its elementary form restricts the model to a linear, isotropic, local and often non-dispersive medium and as such is more than a bridge between two variable families. The constitutive map is therefore simultaneously an architectural component of the theory and a modelling restriction.

Approximations and idealisations should be distinguished. An approximation is ordinarily controlled by a regime:

$$\sin \theta \approx \theta \quad (|\theta| \ll 1).$$

An idealisation introduces a deliberately unreal condition:

massless string, frictionless surface, point particle.

Both simplify, though they do so differently. An approximation asserts closeness within a domain while an idealisation replaces the target with a limiting or fictitious object whose usefulness depends on what it preserves. Restriction therefore answers a different question:

What behaviour is permitted within the model?

Specification: which particular problem?

A specification parse provides the concrete information required to select one member from a family of restricted models. It includes geometry, parameter values, coordinate systems, gauges, initial conditions and boundary conditions. Typical specification questions include:

- What is the geometry?

- What are the material and interaction parameters?
- Which coordinate system makes the structure visible?
- Which gauge fixes representational freedom?
- What is known initially?
- What is imposed at the boundary?
- Which quantities are controlled experimentally?

For a second-order equation such as Newton's

$$m\ddot{x} = -kx,$$

the theory and constitutive choice still define a family of motions while initial data

$$x(0) = x_0, \quad \dot{x}(0) = v_0$$

select a particular trajectory.

Gauge choices occupy an especially instructive position. They do not change the electromagnetic fields, yet they alter the equations presented for the potentials. So for example, the Coulomb gauge,

$$\nabla \cdot \mathbf{A} = 0,$$

and the Lorenz gauge,

$$\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0,$$

specify different representatives of the same physical field configuration. A gauge condition is a specification of representational redundancy and not an empirical law about the world. Confusing those statuses is precisely the sort of error the two-axis analysis of Paper III was designed to expose. Specification then answers the question:

Which member of the model family is being solved?

Reducing different kinds of freedom

The three parses in short reduce different spaces of possibility.

Selection reduces the space of represented entities and variables:

$$\mathcal{T} \longrightarrow \mathcal{S} \subset \mathcal{T},$$

where $\mathcal{T}$ denotes the theory's admissible architecture and $\mathcal{S}$ the selected sector.

Restriction reduces the admissible behaviour within that sector:

$$\mathcal{S} \longrightarrow \mathcal{R} \subset \mathcal{S}.$$

Specification chooses a particular member or solution class:

$$\mathcal{R} \longrightarrow M(\alpha_i, \mathcal{G}, \mathcal{B}, \mathcal{I}),$$

where α_i are parameters, $\mathcal{G}$ geometry or gauge data, $\mathcal{B}$ boundary data and $\mathcal{I}$ initial data.

The sequence is therefore:

$$\boxed{\text{admissible theory}} \rightarrow \boxed{\text{selected sector}} \rightarrow \boxed{\text{restricted model family}} \rightarrow \boxed{\text{specified problem}}.$$

In actual work the order may loop. A boundary condition may reveal that an omitted interaction matters; a numerical instability may expose an invalid approximation; experimental data may force a new constitutive choice. Our taxonomy classifies the logical function of a move, not its temporal order during inquiry.

Worked model I: a block on an inclined plane

The archetypal inclined-plane problem is usually presented as if the diagram already contained the physics whereas the diagram is the endpoint of several modelling decisions.

Selection

The modeler selects:

- the block as the focal body;
- the Earth and plane through their interactions with the block;
- translation of the centre of mass as the retained mechanical degree of freedom;
- gravity, contact force and possibly friction as the included interactions.

Rotation, deformation, sound, heating of the plane, aerodynamic drag and microscopic contact dynamics are excluded. If the Earth is not included inside the system, gravity enters as an external interaction. If block and Earth are included together, gravitational potential energy may be represented internally instead. The statement

$$\sum \mathbf{F} = m\mathbf{a}$$

is therefore not yet a model but merely the dynamical hinge inherited from the theory. The list of applicable forces of contact and gravitational is a selection decision.

Restriction

The block may be restricted to a rigid body form or, indeed just a point mass while the gravitational field may be treated as uniform:

$$\mathbf{g} = \text{constant}.$$

The plane may be idealised as perfectly rigid with friction merely represented by a static inequality,

$$|F_f| \leq \mu_s N,$$

or modelled through kinetic Coulomb friction,

$$F_f = \mu_k N.$$

Each choice changes the model. “A block on a slope” does not identify a unique physical problem until these restrictions are stated. Even the decomposition

$$mg \sin \theta, \quad mg \cos \theta$$

is a choice which relies on a geometric representation in which one axis is parallel to the plane and one is normal to it. These components being one instance of a coordinate expression of a systems of interactions.

Specification

A particular problem requires values or symbols,

$$m, \quad \theta, \quad \mu_s, \quad \mu_k,$$

together with initial conditions such as

$$x(0) = x_0, \quad \dot{x}(0) = v_0.$$

It may also require a controlled-variable declaration as asking how acceleration varies with θ at fixed μ_k differs from asking how the limiting angle varies with μ_s .

For a sliding block with kinetic friction, the completed model is

$$m\ddot{x} = mg \sin \theta - \mu_k mg \cos \theta,$$

or

$$\ddot{x} = g(\sin \theta - \mu_k \cos \theta).$$

The cancellation of m occurs only after selection, restriction and specification have produced the model and is not a free-standing consequence of $F = ma$.

Modelling move	Inclined-plane instance
Selection	Block as focal body; translational degree of freedom; gravity, normal reaction and friction retained.
Restriction	Rigid or point-like block; uniform g ; rigid plane; Coulomb friction; negligible air resistance.
Specification	Angle θ , mass m , coefficients μ_s, μ_k , coordinate axis, and initial data.

Table 1: The apparently elementary inclined-plane problem as a sequence of modelling decisions.

This example also clarifies the concern developed in *Educational Phlogiston*. An equation such as $F = ma$ can be pedagogically overburdened because it suppresses the modelling work that gives F content. The resulting difficulty is more than algebraic load, being the invisibility of the selection and restriction parses that have already been made on the student's behalf.

Worked model II: a charged particle in electromagnetic fields

Consider now a particle of charge q and mass m entering a region containing electric and magnetic fields. The theory supplies Maxwell's equations, particle dynamics and the Lorentz interaction. Several distinct models can be built from this underlying theoretical architecture.

Selection

The standard school model selects:

- one particle with state $(\mathbf{r}, \mathbf{v})$;
- prescribed fields $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{B}(\mathbf{r}, t)$;
- the Lorentz interaction;

- translational motion only.

It excludes:

- the particle's contribution to the source densities;
- radiation emitted by the accelerating charge;
- radiation reaction;
- quantum spin and quantum transitions;
- collisions with residual matter.

The resulting architecture is one-way:

$$(\mathbf{E}, \mathbf{B}) \longrightarrow \mathbf{F} \longrightarrow (\mathbf{r}, \mathbf{v}).$$

Whereas a self-consistent theory would add

$$(\mathbf{r}, \mathbf{v}) \longrightarrow (\rho, \mathbf{J}) \longrightarrow (\mathbf{E}, \mathbf{B}),$$

closing the back-reaction loop. The omission is therefore a selection move: a test-particle model is chosen.

Restriction

The usual Lorentz model assumes classical motion:

$$v \ll c$$

if one also uses

$$\mathbf{p} = m\mathbf{v}.$$

At relativistic speeds the interaction law may retain the same three-force form while the constitutive relation changes:

$$\mathbf{p} = \gamma m_0 \mathbf{v}.$$

A mass spectrometer often adds the restriction

$$\mathbf{E} = 0, \quad \mathbf{B} = \text{uniform and static},$$

within the analysing region. A velocity selector instead imposes crossed uniform fields and seeks trajectories satisfying

$$q\mathbf{E} + q\mathbf{v} \times \mathbf{B} = \mathbf{0}.$$

Neglect of radiation reaction is another restriction of regime being acceptable when radiative losses do not appreciably alter the trajectory over the timescale of interest.

Specification

The problem becomes determinate when one supplies:

$$q, \quad m, \quad \mathbf{E}, \quad \mathbf{B}, \quad \mathbf{r}(0), \quad \mathbf{v}(0),$$

together with the field geometry. For a particle entering perpendicular to a uniform magnetic field,

$$qvB = \frac{mv^2}{r},$$

hence

$$r = \frac{mv}{|q|B}.$$

In relativistic form,

$$r = \frac{p}{|q|B} = \frac{\gamma m_0 v}{|q|B}.$$

The dynamical hinge is stable:

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}.$$

The constitutive completion changes with regime, while the selected interaction and specified geometry determine the orbit.

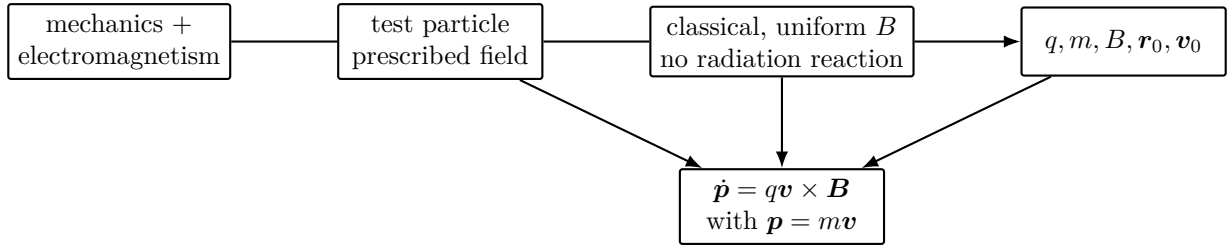


Figure 3: The charged-particle model is produced by successive choices, not by the Lorentz equation alone.

The same theory can produce a mass-spectrometer model, a railgun model or a cyclotron model. What changes is the selection of system and interactions, the restriction to a regime and constitution, and the specification of geometry and conditions.

Worked model III: a thermodynamic cycle

Thermodynamics makes the distinction between theory and model especially clear because its general laws are highly under-determining. The first and second laws constrain possible processes, yet neither supplies an *equation of state*, a process path, reservoir temperatures or device geometry.

Consider a gas undergoing a simple cycle.

Selection

The modeller selects:

- a fixed quantity of gas as the system;
- pressure, volume, temperature and internal energy as state variables;
- heat and work as boundary transfers;
- one or more reservoirs as external sectors.

Microscopic molecular coordinates are not represented. Chemical reactions, phase changes, gravitational stratification and spatial gradients may be excluded.

The system boundary determines the energy account according to the first law:

$$\Delta U = Q - W$$

for the convention in which W is work done by the gas and heat and work classify transfers across the chosen boundary.

Restriction

The gas may be restricted to equilibrium states and ideal-gas behaviour:

$$pV = nRT.$$

The ideal-gas equation is a constitutive closure as the first law alone does not relate p , V and T . Additional restrictions define the process branches:

$$\text{isothermal: } T = \text{constant}, \quad \text{isochoric: } V = \text{constant}, \quad \text{adiabatic: } Q = 0.$$

For a reversible adiabatic ideal-gas branch, $pV^\gamma = \text{constant}$, which contains several restrictions at once: ideal gas, quasistatic path, reversibility and adiabatic insulation.

Specification

A particular cycle requires:

- the amount of gas n ;
- the reservoir temperatures;
- the state points (p_i, V_i, T_i) ;
- the sequence and orientation of process branches;
- heat capacities or γ ;
- any efficiency definition and sign convention.

The area enclosed in a p - V diagram gives the net work:

$$W_{\text{net}} = \oint p \, dV.$$

The sign follows from the direction of traversal. The same geometric loop can represent an engine or refrigerator depending on orientation and on the placement of heat exchanges.

Modelling move	Thermodynamic-cycle instance
Selection	Fixed gas as system; p, V, T, U retained; heat and work treated as boundary transfers; microscopic states omitted.
Restriction	Equilibrium thermodynamics; ideal-gas constitution; specified process classes such as isothermal or adiabatic; reversible or irreversible regime.
Specification	State points, amount n , reservoir temperatures, heat capacities, path orientation and sign convention.

Table 2: Thermodynamic laws constrain a model but do not construct the cycle without additional choices.

This example exposes the role of the hinge particularly clearly. The first law of thermodynamics organises the energy account, though it does not determine Q , W , the process path or the relation among state variables. The open slots are completed by model-level decisions: equation of state, process restriction, system boundary and endpoint specification.

Coordinates, gauges and symmetries

Some modelling decisions alter the representation while leaving the physical content unchanged. They deserve separate attention because their mathematical visibility can make them appear more physical than they are.

Coordinate choice

Coordinates determine how a model is expressed, not which phenomena exist. Cartesian coordinates are convenient for uniform fields and rectangular boundaries; polar coordinates reveal rotational symmetry; field-aligned coordinates may simplify motion in magnetic systems.

As already discussed, for a block on a slope, choosing axes parallel and perpendicular to the plane converts $\mathbf{F}_g = m\mathbf{g}$ into components $F_{\parallel} = mg \sin \theta$, $F_{\perp} = mg \cos \theta$.

Symmetry Choice

Symmetry can function in two related ways. As a selection move, it suppresses degrees of freedom: spherical symmetry replaces a general field $f(r, \theta, \phi)$ by $f(r)$. As a restriction, it asserts that the retained system occupies a symmetric subfamily of solutions, and this dual role matters. The assumption

$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial \phi} = 0$$

may be justified by the source geometry, imposed as an *idealisation*, or used as an exploratory *ansatz*. The same mathematical simplification can therefore possess different epistemic statuses.

Gauge choice

Gauge freedom differs from physical symmetry because it relates mathematically different descriptions of the same physical state. If

$$\mathbf{A}' = \mathbf{A} + \nabla\chi, \quad \phi' = \phi - \frac{\partial\chi}{\partial t},$$

then the fields remain unchanged:

$$\mathbf{B}' = \mathbf{B}, \quad \mathbf{E}' = \mathbf{E}.$$

A gauge choice specifies one representative from an equivalence class. It is a specification move, not a constitutive law or interaction. The distinction is educationally consequential, for a student who sees

$$\nabla \cdot \mathbf{A} = 0$$

without knowing its gauge status may treat it as another empirical field equation. The model taxonomy and the epistemic taxonomy thus meet here: the modelling function is specification, while the epistemic status is representational convention.

A model ledger

The three-parse framework suggests a compact ledger for making a model explicit. Validation is included as a fourth row because modeling in practice is iterative and so tests the model produced by the other three.

Table 3: A model-construction ledger.

Move	Question	Typical entries
Selection	What is included?	System boundary; bodies and fields; degrees of freedom; internal and external interactions; omitted effects; level of description.
Restriction	How may it behave?	Constitutive law; idealisation; approximation; symmetry; regime; linearity; dissipation; equilibrium; continuum assumption.
Specification	Which case is being solved?	Geometry; parameters; coordinates; gauge; initial data; boundary data; controlled variables; numerical constants.
Validation	How will adequacy be judged?	Observable predictions; uncertainty; residuals; domain checks; comparison with data; sensitivity to assumptions.

A model can fail in correspondingly different ways:

- **selection failure:** a relevant interaction or degree of freedom was omitted;
- **restriction failure:** an approximation or constitutive assumption was invalid;
- **specification failure:** parameters, conditions, geometry or calibration were wrong;
- **solution failure:** the stated model was solved incorrectly;
- **measurement failure:** the comparison between model and world was inadequately represented.

This diagnostic structure is more informative than saying simply that “the model was wrong”.

The relation to Papers I and III

Paper I classified equations by architectural role:

structure, constitution, dynamics, interaction.

The present classification answers a different question. A constitutive equation can be available within a theory while the modeller must still choose which constitution applies. Electromagnetism permits general material response; so choosing $\mathbf{D} = \varepsilon \mathbf{E}$ for a linear isotropic medium is a restriction move. Mechanics permits many interactions; deciding that friction is absent is a selection move. A dynamical equation may require initial data; supplying them is a specification move.

Paper III classified equations by epistemic status:

definition, identity, empirical law, approximation,
idealisation, condition, derived result, ...

Again the axes differ. “Approximation” is an epistemic status, but adopting an approximation is a restriction move. “Boundary condition” is an epistemic type, but supplying a particular boundary condition is a specification move. “Interaction law” has an architectural role, but deciding whether that interaction lies inside the selected model is a selection move.

The three classifications can therefore be layered:

Level	Governing question	Example
Theory architecture	Where can this relation sit?	The Lorentz law is an interaction coupling.
Model construction	How is it instantiated here?	The field is prescribed and back-reaction omitted.
Equation epistemology	What kind of claim is it?	The Lorentz law is an empirical coupling law.

Table 4: Architecture, modelling function and epistemic status provide three distinct descriptions of the same physical statement.

Why the hinge works

The hinge framework holds one relation fixed while students vary the completion appropriate to different cases. Paper I explained this architecturally: the hinge is commonly structural or dynamical, whereas the completion is often constitutive or interactional. Paper III explained the epistemic asymmetry: the hinge may be definitionally or structurally authoritative while the completion carries empirical, regime-dependent or conditional content. A hinge relation belongs to the reusable architecture of the theory:

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}, \quad v = f\lambda, \quad \Delta U = Q - W.$$

While the case-specific completion belongs to the model:

$$\mathbf{F} = -k\mathbf{x}, \quad v = \sqrt{\frac{T}{\mu}}, \quad pV = nRT,$$

together with system boundaries, geometries, approximations and conditions.

For the force hinge we have,

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}$$

leaves at least two questions open:

Which interactions have been selected?

and

How are those interactions modelled?

For the wave hinge,

$$v = f\lambda$$

leaves open the constitutive or medium model that fixes v .

For the energy hinge,

$$\Delta U = Q - W$$

leaves open the system boundary, process path and equation of state.

The student who can identify the open slot is recognising a modelling decision that has not yet been made. That is closer to professional physical reasoning than either “find the right equation” or “understand the concept” treated in isolation.

Implications for physics teaching

A model-centred problem should make at least some modelling decisions visible rather than silently embedding all of them in the diagram and wording.

A compact classroom prompt can ask students to complete four statements:

1. **System:** What is inside the model, and what is external?
2. **Restrictions:** Which idealisations, approximations, constitutive laws and regimes are being used?
3. **Specifications:** Which geometry, parameters and conditions define this case?
4. **Open slot:** Which further statement is required before the equations determine a prediction?

This can be represented as

$$\boxed{\text{selected system}} + \boxed{\text{restricted behaviour}} + \boxed{\text{specified case}} \\ + \boxed{\text{governing architecture}} \longrightarrow \boxed{\text{solvable model}}.$$

The purpose is not to require students to catalogue every negligible effect. The goal is discrimination: to identify the assumptions that materially determine the form and validity of the model.

A useful transfer task changes one modelling move at a time:

- retain the same system but relax an approximation;
- retain the same theory and restriction but move the boundary;
- retain the same equations but change the initial conditions;
- retain the same interaction but change the constitutive law;
- retain the same physical fields but change gauge or coordinates.

Such contrasts show students where a result comes from. They also distinguish physical change from representational change.

Validation and revision

Selection, restriction and specification construct a model. Scientific modelling continues by producing consequences and comparing them with the target system.

A minimal cycle is:

$$\text{construct} \longrightarrow \text{solve} \longrightarrow \text{compare} \longrightarrow \text{diagnose} \longrightarrow \text{revise}.$$

Hestenes’s modelling account includes validation as a central practice rather than an optional final check [4]. Model-based laboratory research similarly emphasises construction, prediction, interpretation, identification of limitations and revision [8].

The three-move taxonomy helps locate revisions. If a projectile model systematically underpredicts deceleration, the problem may be selection: drag was omitted. If a pendulum model fails at large amplitude, the problem is restriction: the small-angle approximation was exceeded. If a thermal prediction is offset by a constant factor, the problem may be specification: a heat capacity, calibration or amount of substance was wrong.

Validation is therefore not a binary judgement of true or false. It tests whether the selected, restricted and specified model is adequate for its intended purpose and domain.

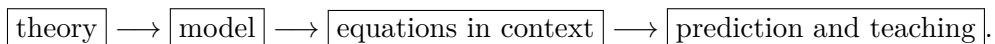
The four-paper hierarchy

The programme can now be stated as a sequence of four questions:

Paper	Object	Question
I	Theory	What kinds of quantities exist, and what architectural roles do their equations play?
II	Teaching	Which module can be held invariant while students vary the case-specific completion?
III	Equation	What kind of claim is this equation making, and what authority does it possess?
IV	Model	Which instantiation of the theory applies here, under which selections, restrictions and specifications?

Table 5: The four papers classify different objects and answer different questions.

There is also a logical order:



Theory provides the space of admissible descriptions. Modelling selects and constrains one such description. Equations acquire their concrete role and authority inside that model. Teaching can then preserve one module while varying another so that the architecture becomes visible.

Conclusion

Physicists rarely solve theories in the abstract. They solve models constructed from theories. The construction requires decisions that are often compressed into phrases such as “assume”, “neglect”, “take” or “let”. This paper has separated those decisions into three kinds.

Selection fixes what the model contains: system, environment, interactions and degrees of freedom. Restriction fixes how the retained components may behave: constitution, idealisation, approximation, symmetry and regime. Specification fixes the particular case: geometry, parameters, coordinates, gauge, boundary conditions and initial data.

The distinction is analytically useful because each move can fail independently, each changes a different space of possibilities, and each has a different relationship to the architectural and epistemic classifications developed elsewhere in the series.

The block on a slope, charged particle and thermodynamic cycle show that even elementary problems are not direct applications of laws. They are carefully restricted constructions and their familiar equations become solvable only after the system has been selected, the behaviour restricted and the case specified.

The hinge framework can now be located precisely. The hinge relation is inherited from the theory; the completion belongs to the model and to identify what is missing is to identify which modelling decision remains open.

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